



Properties and Applications of Metals

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Aims of session



- Discuss classification of metals and their properties
- Establish relationships between microstructure and properties
- Discuss the some applications of different classes of metals

Properties and Applications of Metals

Book: Chapter 13

Metals are the best example
of a man-made material

The use of metals sets
mankind apart from the
rest of nature



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Metals

Metals have defined the advance of technology for 1000's years

Ancient Greece (Bronze) →

Ancient Britain (Iron) ↓



↑ Now (Nano-structured metal)



← 1500 Steel

1779 Cast Iron

→

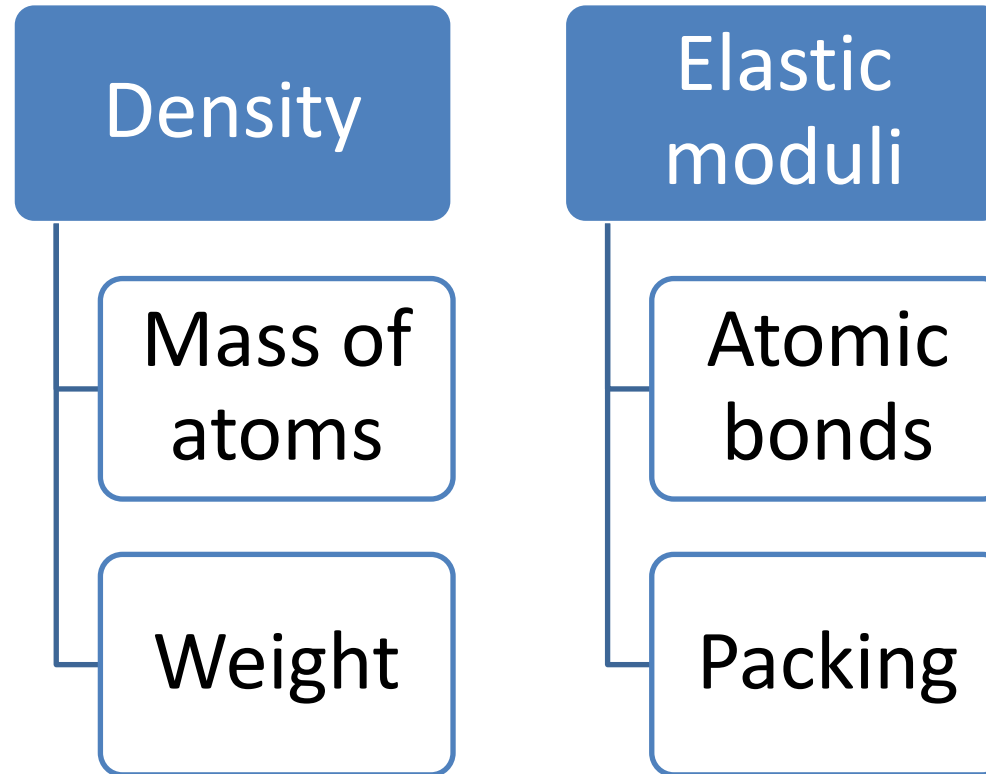


What makes metals great materials?



- Send answers in WeCom

What makes metals great materials?



Metals have useful properties including **strength, ductility, high melting points, thermal and electrical conductivity, and toughness.**

Classification of Metals



Metals are divided into 2 families or classes

Ferrous metals (where the main element is iron, Fe)

- Carbon steel
- Alloy steel
- Cast iron

Non-Ferrous metals (metals where the main element is **NOT** Fe)

- Aluminium alloys
- Copper alloys
- Titanium alloys
- Nickel alloys
- Others (refractory metals, noble metals,)

Ferrous Metals

- Iron – C alloys (steel) are very widely used ($\sim 1.6 \times 10^9$ tonnes/year)
- **China makes 50% of all the steel in the world!**

Construction – high strength



Tool & spring
steels
Very high
strength

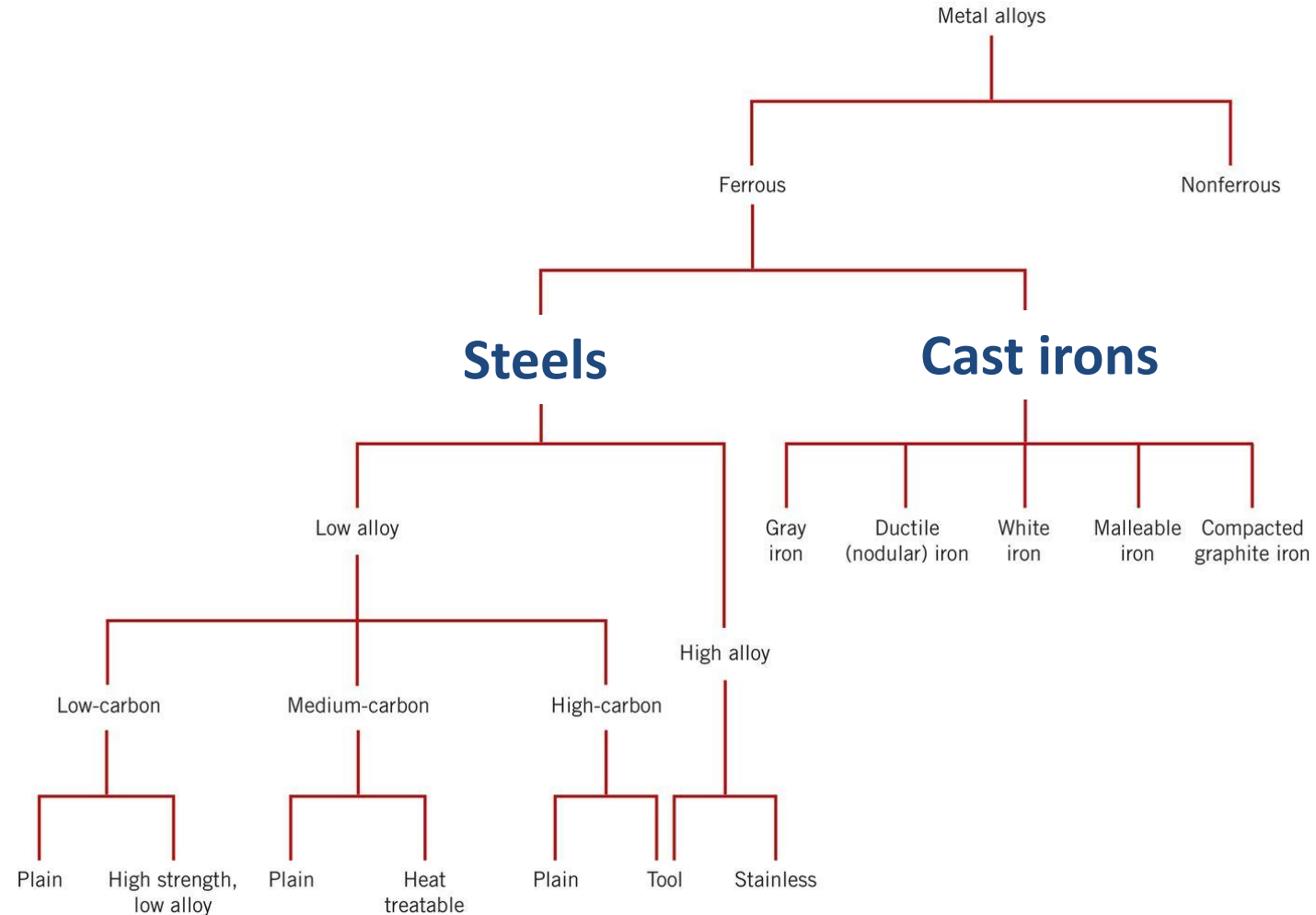


Stainless steel –
good corrosion
resistance

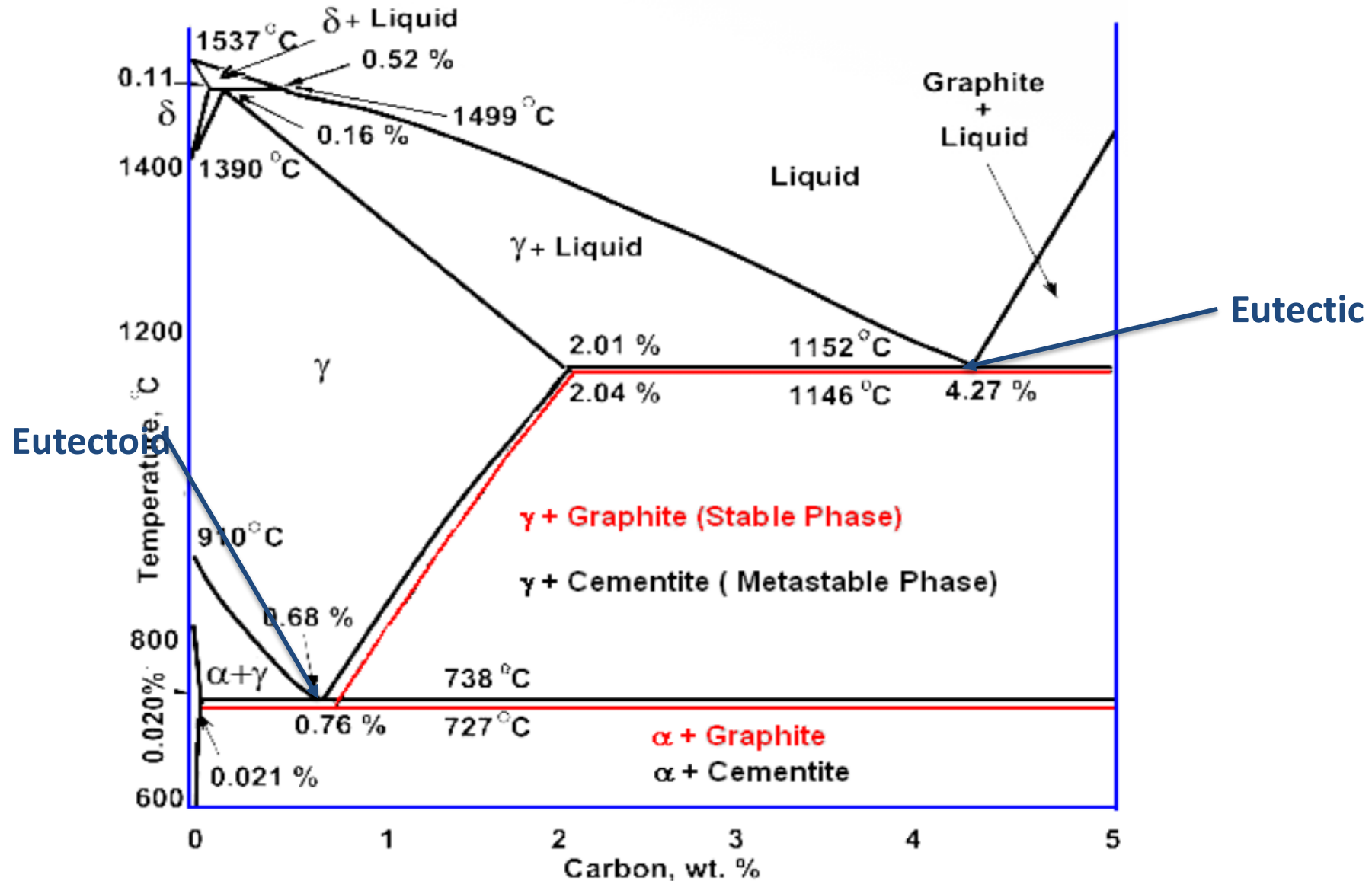
Ferrous Metals



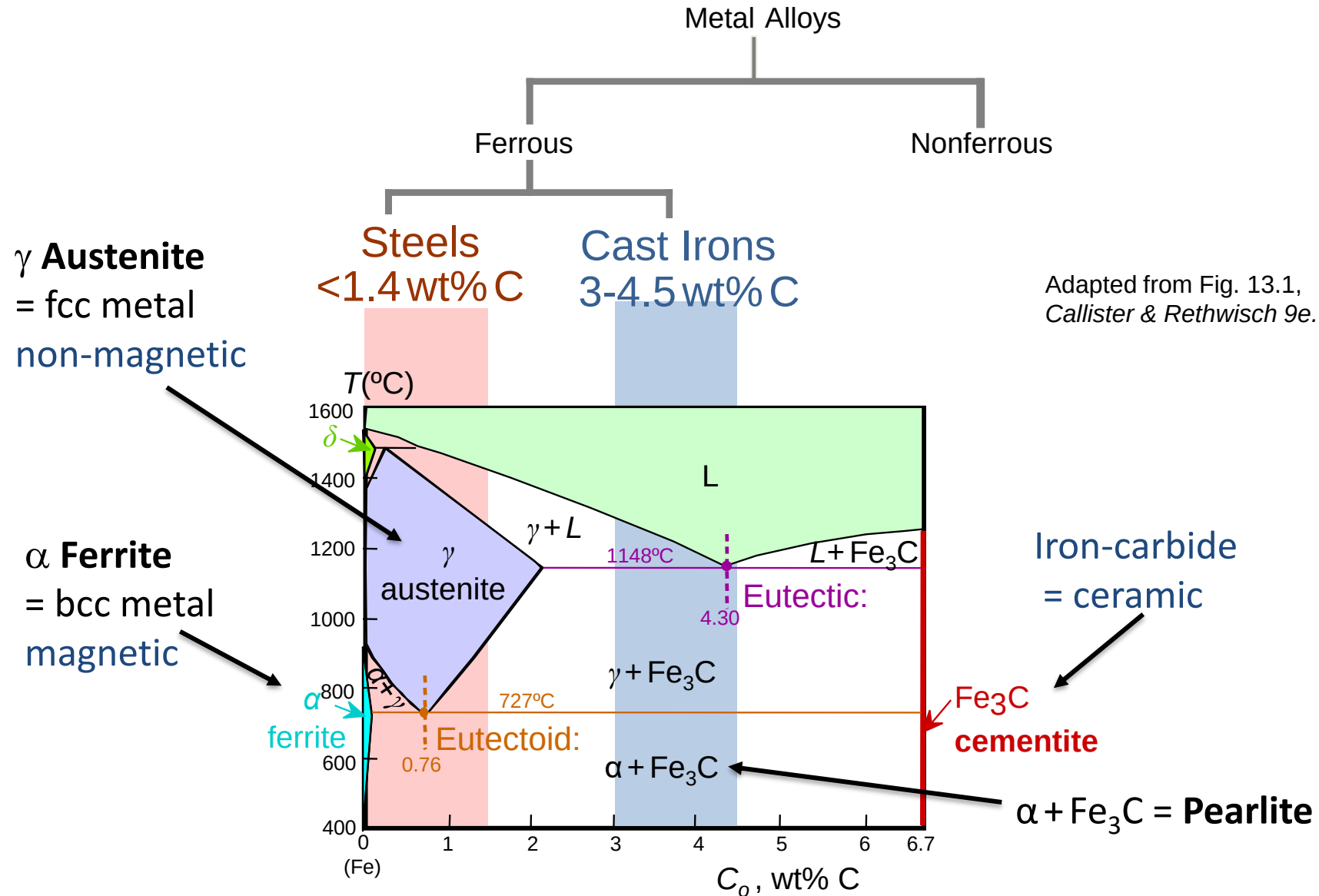
Ferrous Metals are sub-divided into categories:



Fe – C phase diagram

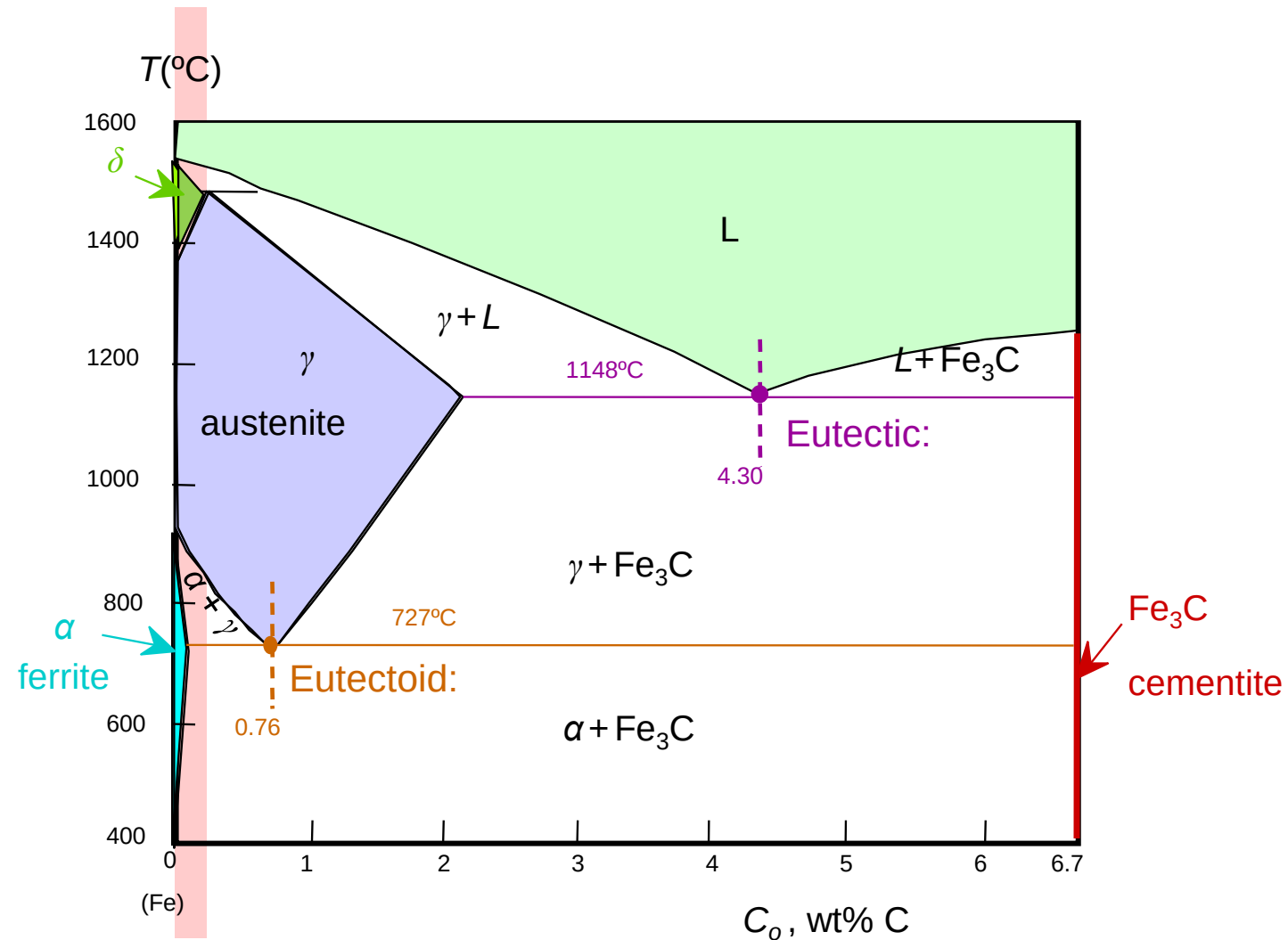


Fe – C phase diagram



Low carbon steels

Low carbon steel = Carbon < 0.25wt%



Low carbon steels

Microstructure:

Composed of α ferrite (soft iron) and pearlite ($\alpha + \text{Fe}_3\text{C}$)

Transformed from $\alpha + \gamma$ austenite on cooling below 727°C

Mechanical Properties:

- Elastic modulus $E = 200 \text{ GPa}$ (all steels = metallic bond of Fe)
- Yield strength $\sigma_y = 275 \text{ MPa}$
- Tensile strength $\sigma_{\text{UTS}} = 415\text{-}500 \text{ MPa}$
- Ductility = 25% EL
- Machinable, deformable, weldable

Disadvantage:

Corrosion – electronegative in the galvanic series = oxidation

Question 1



Question for:

Materials Science and Engineering students

WeCom group: QMES2021M

Q1. Why do low carbon steels NOT respond to heat treatment?

- A. Their melting point is too high
- B. Their strength is too low
- C. There is not enough carbon to transform to martensite
- D. They are magnetic materials

Low carbon steels

Microstructure:

Composed of α ferrite (soft iron) and pearlite ($\alpha + \text{Fe}_3\text{C}$)

Do not respond to heat treatment

Strengthening by cold working – dislocation structures

Low carbon **alloy** steels (high strength-low alloy steel):

Alloy with Cu, V, Ni, Mo

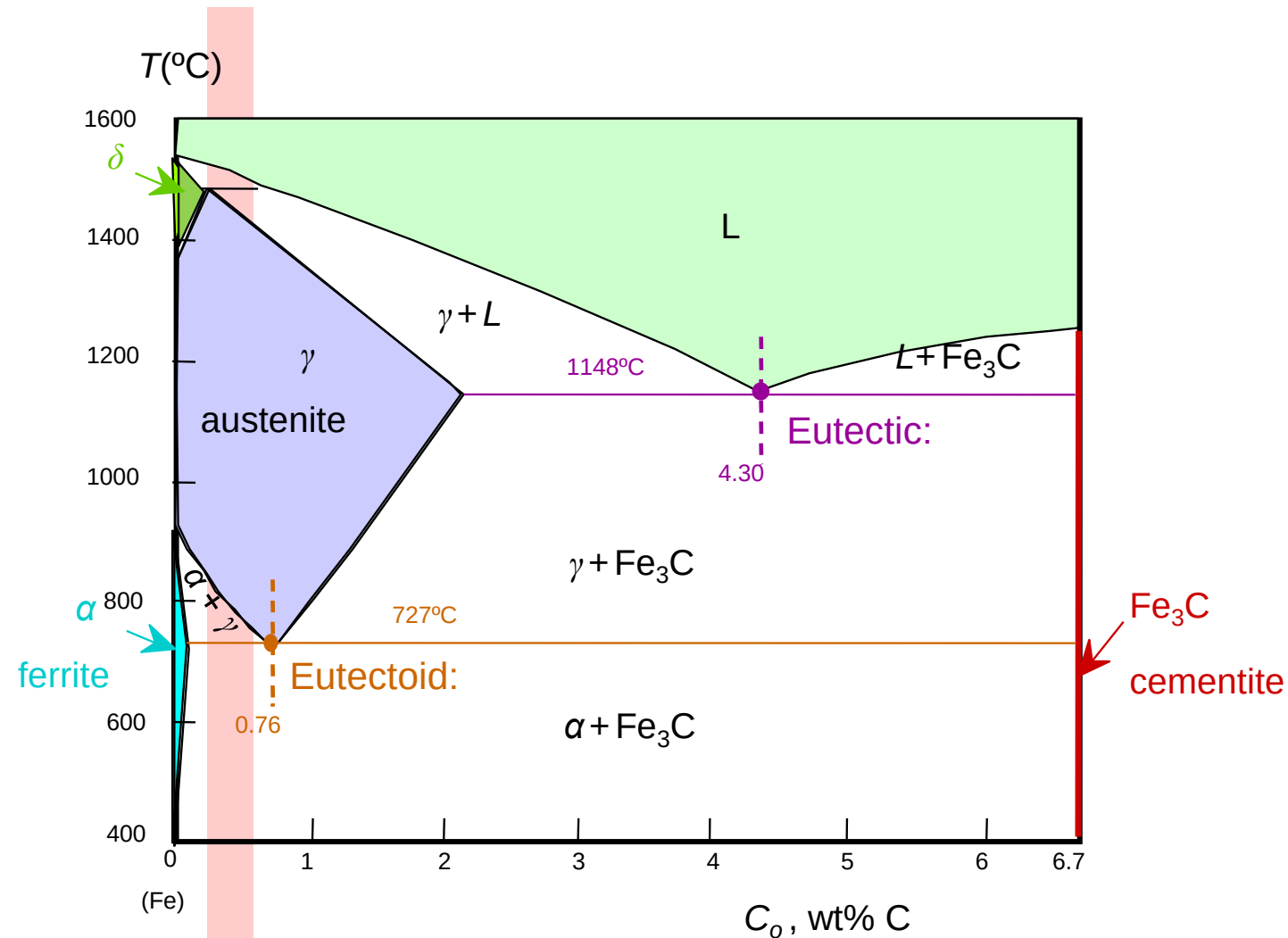
- Yield strength $\sigma_y = 290\text{-}550$ MPa
- Tensile strength $\sigma_{\text{UTS}} = 435\text{-}650$ MPa
- Ductility = 15-20% EL

Applications:

Car body panels, structural beams, construction, pipes
bridges, high rise buildings, pressure vessels

Medium carbon steels

Medium carbon steel = $0.25\text{wt}\% < \text{Carbon} < 0.6\text{wt}\%$



Medium carbon steels

Microstructure:

Composed of α ferrite (soft iron) and pearlite ($\alpha + \text{Fe}_3\text{C}$)

Transformed from γ austenite on cooling below 727°C

Heat treat – austenise, quench, temper – to form Martensite

Mechanical Properties:

- Elastic modulus $E = 200 \text{ GPa}$ (all steels – metallic bond of Fe)
- Yield strength $\sigma_y = 400\text{-}800 \text{ MPa}$
- Tensile strength $\sigma_{\text{UTS}} = 600\text{-}1200 \text{ MPa}$
- Ductility = 10-30% EL
- Wide range of hardenability

Applications:

Bolts, shafts, hammers, knives

Medium carbon **alloy** steels:

Alloy with Ni, Cr, Mo

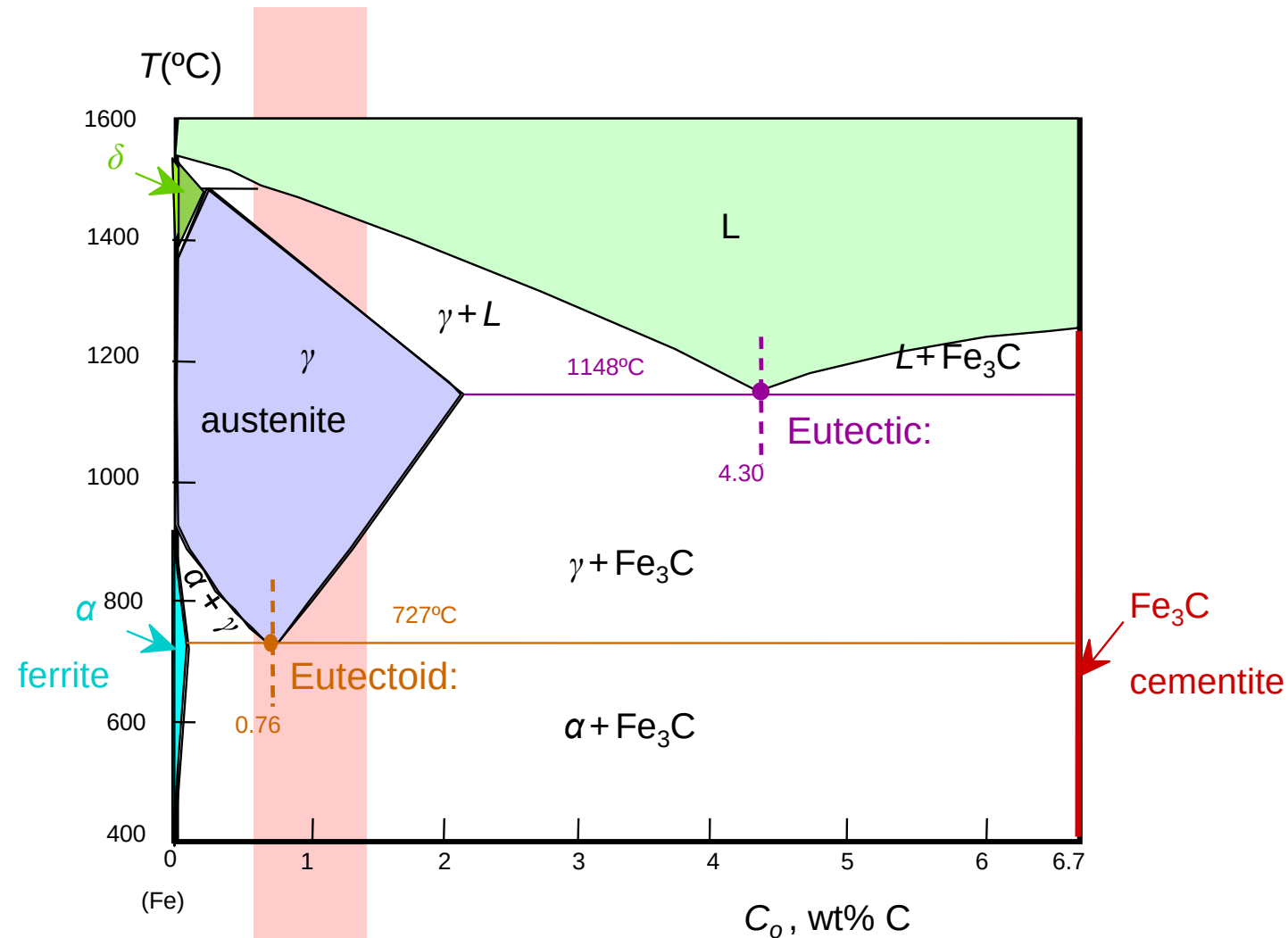
- Yield strength $\sigma_y = 700-1860$ MPa
- Tensile strength $\sigma_{UTS} = 780-2000$ MPa
- Ductility = 4-20% EL

Applications:

Springs, hand tools, bushings, shafts, pistons, gears
(high wear resistance)

High carbon steels

High carbon steel = $0.6\text{wt}\% < \text{Carbon} < 1.4\text{wt}\%$



High carbon steels



Microstructure:

Composed of pearlite ($\alpha + \text{Fe}_3\text{C}$) and / or Martensite

Heat treat – austenise, quench, temper – to form Martensite

Alloy with Cr, V, W, Mo to form (Cr_{23}C_6 , V_4C_3 , WC)

hard, wear resistant carbide particles in steel

Applications:

Hard tools and machinery (**tool steel**)

Drills, saws, punches, dies, cutting blades

Stainless steels



Reduce corrosion by alloying with > 11%Cr (low carbon):

Ferritic stainless steel 11-25wt% Cr (magnetic)

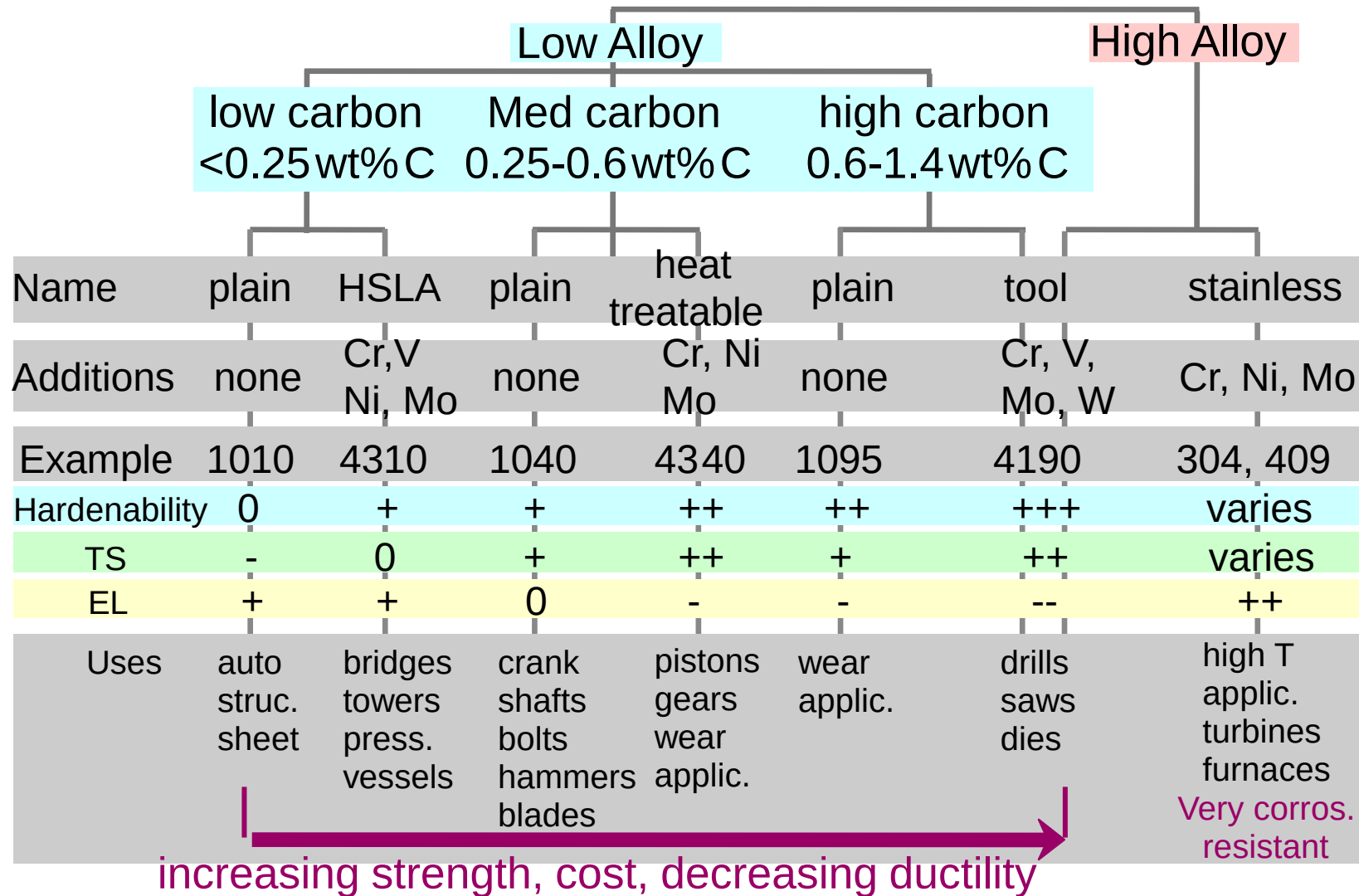
Austenitic stainless steels 17-19wt% Cr + 9-12wt%Ni (non-magnetic)

- Elastic modulus $E = 190$ GPa
- Yield strength $\sigma_y = 200-275$ MPa
- Tensile strength $\sigma_{UTS} = 380-500$ MPa
- Ductility = 20-40% EL

Applications:

Combustion chambers, chemical processing, valves
Chemical and food processing

Summary of steels



Typical mechanical properties of several metals and alloys in an annealed state

<i>Metal Alloy</i>	<i>Yield Strength, MPa (ksi)</i>	<i>Tensile Strength, MPa (ksi)</i>	<i>Ductility, %EL [in 50 mm (2 in.)]</i>
Aluminum	35 (5)	90 (13)	40
Copper	69 (10)	200 (29)	45
Brass (70Cu–30Zn)	75 (11)	300 (44)	68
Iron	130 (19)	262 (38)	45
Nickel	138 (20)	480 (70)	40
Steel (1020)	180 (26)	380 (55)	25
Titanium	450 (65)	520 (75)	25
Molybdenum	565 (82)	655 (95)	35

Table_6-2 Callister and Rethwisch, 7e

Poll –
Using the
table in the
previous slide,
answer the
following
questions



Q1. Which of these metals will yield first, Al or Mo?



Q2. Which of these metals is better at standing plastic deformation, Cu or Fe?



Q3. Which of these metals is more brittle, Ni or Ti?

Answers



- Q1. Aluminium. The value of yield strength is 35 MPa compared to 565 MPa for Mo, which means Al will experience plastic deformation under a low load or stress.
- Q2. Iron. The value of tensile strength is 262 MPa compared to 200 MPa for Cu. This indicates that a higher stress is required to fracture a material made of Fe under tension.
- Q3. Titanium. The ductility value for Ti is 25% EL compared to the value of Ni (40% EL). This suggests that Ti will experience higher plastic deformation at fracture.

Additional work – On-line learning



- Watch the following videos in QMPlus
 - Introduction to metallic glass
 - Application of metallic glass
- Complete the self-assessment exercise for each video